Lecture 14: Mild Solution Analysis and Rough Integral Estimates

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Good morning, class. Let us begin today's lecture on Stochastic Partial Differential Equations. We will go through the Lecture 14 notes in detail, focusing on the functional estimates and the rough path integration techniques that arise in our evolution equations.

1. Mild Decomposition of the Process

2.1. Decomposition Formula

We begin by placing the equation in the mild solution framework for our stochastic partial differential equation. The solution process $u(t,x)$ is decomposed into three components:

$$u(t,x) = v(t,x) + X_t(x) + e^{-t}P_t(u_0 - X_0).$$

In words, the solution at time t and position x is written as the sum of a smoother remainder term $v(t,x)$, the principal rough noise component $X_t(x)$, and the damped heat semigroup applied to the shifted initial data. The interpretation is that the solution is decomposed into a regular part, a rough stochastic part, and the propagated initial condition. The purpose of this decomposition is to isolate the rough part of the solution and reduce the remaining analysis to the more regular term v.

2.2. Mild Form and Operator Splitting

Next, we write the integral equation for the remainder process $v(t,x)$. It is given by the sum of two integral operators:

$$v(t,x) = \int_0^t e^{-(t-s)}P_{t-s}\big(g(u(s,\cdot))\partial_x v(s,\cdot)+f(u(s,\cdot))\big)(x)\,ds+\int_0^t\int_0^{2\pi} e^{-(t-s)}P_{t-s}(x-y)g(u(s,y))\,dX_s(y)\,ds.$$

We label the first time-integral term by $M^{(1)}_{T,X}v(t,x)$ and the second space-time term by $M^{(2)}_{T,X}v(t,x)$. Explicitly,

$$M^{(1)}_{T,X}v(t,x) = \int_0^t e^{-(t-s)}P_{t-s}\big(g(u(s,\cdot))\partial_x v(s,\cdot) + f(u(s,\cdot))\big)(x)\,ds,$$

In words, $M^{(1)}_{T,X}v(t,x)$ is the deterministic contribution to the remainder process: it accumulates, over all earlier times s, the drift term $f(u(s,\cdot))$ and the transport term $g(u(s,\cdot))\partial_x v(s,\cdot)$, and then propagates them from time s to time t by the damped heat semigroup $e^{-(t-s)}P_{t-s}$. The interpretation is that $M^{(1)}_{T,X}$ collects the regular deterministic forcing generated by the nonlinearity.

and

$$M^{(2)}_{T,X}v(t,x) = \int_0^t\int_0^{2\pi} e^{-(t-s)}P_{t-s}(x-y)g(u(s,y))\,dX_s(y)\,ds.$$

In words, $M^{(2)}_{T,X}v(t,x)$ is the rough forcing contribution: it integrates the explicitly weighted term $g(u(s,y))$ against the spatial rough signal $dX_s(y)$ over $y \in [0, 2\pi]$, and then propagates the result from time s to the point (t,x) by the damped heat semigroup kernel $e^{-(t-s)}P_{t-s}(x-y)$. The interpretation is that $M^{(2)}_{T,X}$ is the genuinely rough contribution driven by the irregular spatial signal. The purpose of this splitting is to treat the deterministic and rough contributions by different estimates later in the proof.

2. Two Main Steps of the Analysis

Our analysis proceeds in two main steps.

2.1. Step 1: The First Estimate

Our first objective is to establish a rigorous estimate for the operator $M^{(1)}_{T,X}$. We consider the supremum over times $t \le T$ of the C^1 norm of $M^{(1)}_{T,X}v(t)$ and obtain

$$\sup_{t\le T} \|M^{(1)}_{T,X}v(t)\|_{C^1} \le C_k\big(1 + \|X\|_\alpha\big).$$

This says that the maximal C^1 norm up to time T is bounded by a constant C_k times $1+\|X\|_\alpha$. As in the notes, this estimate relies on the classical theory of evolution equations, and the dependence on the Hölder exponent α enters through the rough path norm. The purpose of Step 1 is to show that the deterministic part of the mild formulation already satisfies the required C^1 bound.

2.2. Step 2: The Main Ingredient

Moving to Step 2, our goal is to control the second operator $M^{(2)}_{T,X}$. Although $M^{(2)}_{T,X}$ does not contain an explicit spatial derivative in its definition, we need to estimate its spatial derivative in order to control $M^{(2)}_{T,X}$ in the required C^1-type norm. To do this, we introduce the main ingredient for bounding the spatial derivatives of the rough convolution term. We restrict our regularity parameter α such that $\alpha \in (\frac{1}{3}, \frac{1}{2})$. Under this assumption, we obtain the following inequality:

$$\left|\partial_x \int_0^{2\pi} e^{-t}P_t(x-y)Y_y\, dX_s(y)\right| \le Ct^{\alpha/2-1}\|Y\|_{X,\alpha}\, |||\mathbb{X}|||_\alpha. \tag{1}$$

In spoken terms, the absolute value of the spatial derivative of the rough convolution integral over the domain $[0, 2\pi]$ is bounded above by a constant C, times the time-singular factor $t^{\alpha/2-1}$, times the controlled rough path norm $\|Y\|_{X,\alpha}$, times the rough path norm $|||\mathbb{X}|||_\alpha$. This estimate holds uniformly with respect to the spatial variable x on the interval $[0, 2\pi]$. The interpretation is that the singular spatial convolution still has enough spatial regularity to be controlled in the rough-path framework. The purpose of Step 2 is to show that the rough spatial convolution in $M^{(2)}_{T,X}$ also satisfies the required C^1-type control, despite the singular behavior of the rough forcing.

3. Context: Classical Itô Isometry vs Rough Path Integration

3.1. Classical Itô Isometry

We then transition to the stochastic integration properties. More precisely, the left-hand side of (**??**),

$$\left|\partial_x \int_0^{2\pi} e^{-t} P_t(x-y) Y_y \, dX_s(y)\right|,$$

is not itself justified by the Itô isometry, because it is a spatial rough integral rather than a classical stochastic integral in time. The Itô isometry is mentioned here only as the classical comparison principle. The foundational equation for this isometry is:

$$\mathbb{E}\left(\left(\int_0^s dW_r\right)^2\right) = \mathbb{E}\left(\int_0^s dr\right).$$

This isometry guarantees convergence for classical stochastic integrals with respect to Brownian motion. In contrast, the convergence needed for (**??**) is provided by rough path estimates. You will also see a marginal note reading "Feynman? need to read up", which serves as a reminder to investigate the specific theoretical connections between these measure convergences and Feynman path integrals. The purpose of mentioning Itô isometry is to contrast the classical stochastic setting with the rough spatial setting used later in the lecture.

3.2. Rough Path Definition of the Spatial Integral

Finally, we must precisely define the spatial rough integral $\int_0^{2\pi} Z_y \, dX(y)$, that is, a rough path integral over space. The integral follows the formal limit:

$$\int_0^{2\pi} Z_y \, dX(y) := \lim_{|P|\to 0} \sum_{[x,y]\in P} \big(Z_x(X_y - X_x) + Z'_x \, \mathbb{X}(x,y)\big).$$

This means the rough integral is the limit, as the partition mesh size $|P|$ approaches zero, of the sum over partition intervals $[x,y]$. The summand consists of the classical Riemann–Stieltjes term $Z_x(X_y - X_x)$ and the second-order rough path correction term $Z'_x \, \mathbb{X}(x,y)$, where $(X, \mathbb{X})$ is the rough path lift of X and $\mathbb{X}$ is its second-level component.

Here, X is the rough driving signal, and Z is the function or path that we want to integrate against X.

The critical observation concluding these notes is that this formulation does not require Z_x to be differentiable. Therefore, even if the classical derivative of Z_x does not exist, the rough path framework bypasses this limitation and allows us to maintain mathematical rigor throughout the integration process. The purpose of this formulation is to define the spatial integral rigorously even when classical differentiability is unavailable.

4. Integrand Properties and Rough Bounds

4.1. The Gubinelli Derivative and the Remainder

Let us resume our analysis by looking at the specific properties of the integrand process, and how we handle the lack of classical differentiability. We note that we can use a bounded process which

does not change the term Z'_x. Here, Z'_x is the Gubinelli derivative, and the relationship it must satisfy is given by the equation:

$$R_{x,y} = Z_y - Z_x - Z'_x(X_y - X_x).$$

In words, the remainder term $R_{x,y}$ is defined as the difference between the process Z evaluated at y and x, minus the Gubinelli derivative Z'_x multiplied by the spatial increment of the rough path $(X_y - X_x)$. The interpretation is that $R_{x,y}$ measures exactly how far Z is from being a first-order expansion along the rough signal X.

I must emphasize, as I have annotated in the margins, that this Gubinelli derivative is "not uniquely defined" and is "not diff.", meaning it is not classically differentiable. This conceptual framework only makes mathematical sense if the remainder term $R_{x,y}$ is 2α-Hölder continuous. Specifically, we require the Hölder norm of the remainder to satisfy the condition:

$$\|R\|_{2\alpha} < \infty.$$

Stated plainly, the 2α-Hölder norm of the remainder must be strictly bounded, which ensures the higher-order 3α Hölder term effectively vanishes in the limit. The purpose of this condition is to ensure that the remainder is higher order, so the controlled expansion is mathematically meaningful.

4.2. Contrast with Classical Derivatives

To draw a sharp contrast with classical calculus, we recall the standard definition of a derivative of one function with respect to another:

$$\frac{dg}{df}(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{f(x+h) - f(x)}.$$

Verbally, the classical derivative of f with respect to g is the limit as the perturbation h approaches zero of the difference quotient of f divided by the difference quotient of g. Our rough path framework circumvents the need for this limit to exist in the traditional sense. The interpretation is that rough paths replace a classical derivative-based expansion by a controlled expansion with a Hölder remainder. The purpose of this comparison is to show exactly what rough path theory replaces from classical calculus.

4.3. A Fundamental Rough Integral Bound

This leads us to a fundamental bound for the rough integral, which is written as:

$$\left\| \int_0^{\cdot} (Z_y - Z_0)\, dX_y \right\|_\alpha \leq C\big(\|X\|_\alpha \|R\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Z'\|_\alpha\big).$$

Expressed in words, the α-Hölder norm of the rough integral is less than or equal to a constant C multiplied by the sum of two products: the α-Hölder norm of the path X times the 2α-Hölder norm of the remainder R, plus the 2α-Hölder norm of the second-order component $\mathbb{X}$ times the α-Hölder norm of the Gubinelli derivative Z'. This ensures the 3α component converges properly. The interpretation is that the rough integral is controlled by the roughness of the driver and by how well the integrand is controlled by that driver. The purpose of this estimate is to give a quantitative bound for the rough integral that can be used later in the SPDE argument.

5. Estimates for $M^{(2)}_{T,X}$ and Rough Path 2.0

5.1. The Heat Semigroup Derivative Estimate

This subsection refines the Step 2 discussion in Section 2.2: there we stated the strategy for controlling $M^{(2)}_{T,X}$, while here we derive the more detailed spatial-derivative estimate that is used to bound $M^{(2)}_{T,X}$ in the required C^1-type norm. Applying this to the partial differential equation component, we derive an estimate for the spatial derivative term involving the heat semigroup P_{t-s}. The inequality is:

$$\left|\partial_x \int_0^{2\pi} P_{t-s}(x-y)Y_y\, dX(y)\right| \leq C_K(1+|||X|||)^3 \underbrace{\left(1+s^{(3-2\alpha)/2}\right)(t-s)^{\alpha/2}}_{\sim (t-s)^{\alpha/2-1}}.$$

In spoken terms, the absolute value of the spatial derivative of the convolution integral over the domain 0 to 2π is bounded by a constant C_K times the quantity one plus the triple-bar norm of the rough path X raised to the third power, multiplied by the full factor $\left(1+s^{(3-2\alpha)/2}\right)(t-s)^{\alpha/2}$, which is indicated here as behaving like $(t-s)^{\alpha/2-1}$ for the singular derivative estimate. The interpretation is that differentiating the heat kernel creates the singular time factor, but the overall bound is still explicit enough to integrate in time. The purpose of this estimate is to control the spatial derivative of the rough convolution term that appears in $M^{(2)}_{T,X}$.

5.2. Origin of the $(t-s)$ Singularity

To understand precisely where this $(t-s)$ singularity originates, we make explicit that the spatial derivative in the C^1-estimate falls on the heat kernel. Formally,

$$\partial_x \int_0^{2\pi} P_{t-s}(x-y)Y_y\, dX(y) = \int_0^{2\pi} \partial_x P_{t-s}(x-y)Y_y\, dX(y).$$

Thus the relevant object is the spatial derivative of the heat kernel itself. Writing the kernel in the form

$$P_{t-s}(x-y) \sim \frac{1}{\sqrt{t-s}} e^{-\frac{\|x-y\|^2}{2(t-s)}},$$

the derivation is:

$$\partial_x\left(\frac{1}{\sqrt{t-s}} e^{-\frac{\|x-y\|^2}{2(t-s)}}\right) = -\frac{x-y}{t-s}\frac{1}{\sqrt{t-s}} e^{-\frac{\|x-y\|^2}{2(t-s)}}.$$

Reading this equation out loud: taking the partial derivative with respect to the spatial variable x of the heat kernel—which is one over the square root of $t-s$, times the exponential of negative norm $x-y$ squared over two times the quantity $t-s$—yields negative $x-y$ over $t-s$, multiplied by the original heat kernel expression. This is the step in which the derivative is transferred to $P_{t-s}(x-y)$, and it is exactly this computation that produces the singular factor in $(t-s)$. The purpose of this computation is to identify exactly where the singular factor in $(t-s)$ comes from.

Putting these elements together, we arrive at the critical estimate for our second integral operator, $M^{(2)}_{T,X}$. The bound is:

$$\|M^{(2)}_{T,X} v\|_{1,T} \leq C_K(1+|||X|||)^3 \sup_{t\leq T} \int_0^t (t-s)^{\alpha/2-1}\, ds < T^{\alpha/2}.$$

In words, $\|M^{(2)}_{T,X} v\|_{1,T}$ is bounded by $C_K(1+|||X|||)^3$ times the time integral $\int_0^t (t-s)^{\alpha/2-1}\, ds$. Since $\alpha > 0$, this integral is bounded by a constant multiple of $T^{\alpha/2}$, so the contribution stays finite. The interpretation is that the singularity in time is mild enough that the full operator remains well defined. The purpose of this bound is to show that the singular time factor is still integrable, so $M^{(2)}_{T,X}$ stays finite in the required norm.

5.3. Rough Path 2.0 and Spatial Regularization

Finally, we transition to the topic I have labeled "Rough path 2.0", where we consider a random field with a spatial parameter. We introduce the regularized stochastic partial differential equation model:

$$du_\varepsilon = \partial^2_{xx} u_\varepsilon\, dt + f(u_\varepsilon)\, dt + g(u_\varepsilon)\partial_x u_\varepsilon + \sigma \circ Q_\varepsilon\, dW_t.$$

To vocalize this equation: the differential of the regularized solution u_ε equals its second spatial derivative with respect to x times dt, plus a reaction term $f(u_\varepsilon)$ times dt, plus a transport term $g(u_\varepsilon)$ times the first spatial derivative of u_ε, plus the diffusion coefficient σ composed with a spatial smoothing operator Q_ε integrated against the Wiener process dW_t.

This spatial smoothing operator Q^u_ε acts on a function ξ via convolution. The defining equation is:

$$Q^u_\varepsilon(x) = \int_0^{2\pi} \phi_\varepsilon(x-y) u(y)\, dy \to u(x).$$

In words, $Q^u_\varepsilon(x)$ is defined as the integral from 0 to 2π of the mollifier $\phi_\varepsilon(x-y)$ multiplied by $u(y)$ with respect to y. Here, ϕ_ε is a smooth approximation kernel that averages nearby values and removes small-scale irregularities. As ε approaches zero, this convolution converges back to $u(x)$. This is why the equation is called regularized: the rough field is first smoothed by Q^u_ε before it enters the SPDE. The interpretation is that the mollifier replaces a rough object by a smoother approximation that can be handled before taking a limiting procedure. The purpose of this regularization is to replace the rough equation by smoother approximations that can later be analyzed and then sent to the limit.

6. Discretization and Biological Models

6.1. Discretization and Space-Time White Noise

We now pass to a discrete spatial grid in order to approximate space-time white noise numerically. The notes denote the sampled noise at grid points by $W_t(x)$ and indicate that this is meant to "approximate ST WN." The key technical point is that the resulting rough path approximation depends on how the sampled spatial increments are discretized and assembled.

The purpose of this discretization is to turn the continuous rough model into a form that can be simulated numerically.

6.2. The Nagumo Equation and Nerve Modeling

We construct a coupled system of partial differential equations; we do this to mathematically capture the complex biological process of electrical signal propagation along a nerve axon, incorporating both deterministic biology and random physical fluctuations.

We introduce a model representing a nerve fiber, which the notes depict as a physical domain with an internal "local reaction." The governing evolution equation for this system is the stochastic

FitzHugh–Nagumo equation. The primary equation for the state variable $u(t, x)$ is given by:

$$u_t(t, x) = \partial_{xx} u(t, x) + f(u(t, x)) - v(t, x) + \xi(t, x).$$

Expressed in words, the partial time derivative of the state variable u at time t and spatial position x is equal to the second spatial derivative of u with respect to x, plus a nonlinear local reaction function $f(u)$, minus a secondary recovery variable $v(t, x)$, plus a stochastic noise term $\xi(t, x)$. As annotated in the notes, the addition of the noise term ξ is specifically designed "to explain fluctuation" inherent in the underlying environment. The interpretation is that diffusion, reaction, recovery, and noise all act simultaneously in the nerve model.

The purpose of this model is to combine diffusion, nonlinear reaction, recovery, and noise in one tractable biological system.

6.3. The Bistable Reaction Dynamics

We choose a specific nonlinear cubic form for the local reaction term; we do this to create a bistable system with a built-in threshold, mirroring how a biological neuron requires a stimulus of a certain minimum strength before it triggers an action potential.

The local reaction term $f(u)$ is defined as a specific cubic polynomial to generate this threshold behavior. The equation is:

$$u' = f(u) = u(1 - u)(u - a).$$

In spoken terms, the ordinary derivative of u driven by the reaction is equal to the function $f(u)$, which is the product of u, the quantity one minus u, and the quantity $u - a$, where the parameter a is strictly between zero and one, denoted as $a \in (0, 1)$. The lecture notes provide a sketch of this function, showing it crossing the horizontal axis at three distinct roots: 0, a, and 1. The stable root at 0 is labeled as the "dark site," while the stable root at 1 is labeled as the "white site." The interpretation is that the cubic nonlinearity creates a threshold between two stable states, which is the mechanism behind excitability.

The purpose of this cubic choice is to build in a threshold mechanism and two stable states.

6.4. Traveling Waves

We analyze the spatial profile of the solution over time; we do this for the purpose of understanding how a localized excitation physically travels down the spatial domain as a coherent, self-sustaining structural wave.

When the system is pushed past the unstable threshold a, the dynamics drive the state locally towards the "white site" at 1. The notes illustrate this with a graph of "traveling wave" solutions. The diagram displays multiple wave fronts shifting continuously to the right along the x-axis, with arrows indicating the state pushing upward to 1 at the front and decaying backward. The text clarifies that these waves propagate at a "deterministically constant speed" across the domain.

The purpose of this analysis is to identify how excitation propagates spatially through the medium.

6.5. The Recovery Variable and Bump Solutions

We introduce a secondary, slower differential equation to the system; we do this to ensure that after the spatial region reaches its excited state, it is forced to eventually return to its resting state, thereby creating a localized and transient "bump" of activity rather than a perpetually excited domain.

To manage these "isolated regions" of excitation and prevent the wave from keeping the entire domain at state 1, the notes indicate we "need 2nd variable to push it down." This is the role of the recovery variable $v(t, x)$. Its evolution is governed by the equation:

$$\partial_t v(t, x) = \varepsilon\big(u(t, x) - \gamma v(t, x)\big).$$

Reading this equation aloud: the partial derivative of the recovery variable v with respect to time t is equal to a rate parameter ε multiplied by the difference between the main state variable $u(t, x)$ and a scaled version of the recovery variable, $\gamma v(t, x)$. Because this recovery variable acts as a subtraction in the primary u equation, it eventually forces the excited state back down to 0 once the wave front has passed. As sketched at the very bottom of the notes, this coupled interaction creates a localized, propagating pulse, which is explicitly labeled as a "bump solution." The interpretation is that the slow recovery variable prevents permanent excitation and thereby creates localized pulses instead of a fully activated domain.

The purpose of the recovery variable is to turn a permanent transition into a localized pulse that can travel and then decay.

7. Observation, Estimation, and Bayesian Updating

7.1. The Observation Problem and State Space Formulation

We construct a combined mathematical state vector to capture the complete dynamics of the system; we do this because observing only a single variable of a coupled biological or physical system leads to missing information, which destroys the Markov property and prevents us from writing down a closed-form differential equation.

We begin by considering a physical observation process. As noted at the top of the page, in practice, a technique like "microscopy measures u." There is also an additive component "$+g$" which is tentatively labeled with an arrow as "measurement?" suggesting a measurement operator or noise associated with the observation. However, the underlying system requires us to reconstruct both the observed state u and the unobserved, hidden state v.

To handle this, we formally define our complete state vector as

$$X = [u, v].$$

In words, the state X is composed of the vector containing both the primary variable u and the recovery variable v. The notes highlight a fundamental problem, labeled "Pb": if we try to model the system by observing only u, it is "not markovian if we ignore v." Because the future state of u depends on the current state of v, omitting v means the process loses its memoryless Markov property, and consequently, we "cannot write down PDE" for u alone. The interpretation is that the full state must include every variable needed to make the dynamics closed and Markovian.

The purpose of this state space formulation is to restore a closed Markovian description by including both observed and hidden variables.

7.2. The Frequentist Approach to Estimation

We formulate a direct mathematical inverse problem from our observations; we do this to attempt to retrieve the exact underlying state by treating the governing equations as a deterministic mapping, assuming any measurement noise can simply be averaged out over multiple samples.

We formally state our objective:

$$\text{Est } X \text{ given } Y = G(X, e).$$

Reading this aloud, we want to estimate the true state X given an observation Y, where Y is generated by a function G depending on the state X and some error e. The variable Y is explicitly labeled as the "measurement," and the function G represents our forward model, indicated in the notes to be either an "(S)PDE" or an "(S)ODE"—a stochastic partial or ordinary differential equation. The interpretation is that the inverse problem starts from data and asks which hidden state could have produced them through the forward model.

The notes first outline the "frequentist approach." The standard mathematical procedure here is straightforward: "If G invertible $\to$ solve for X." In spoken terms, if the forward operator G has a well-defined mathematical inverse, we can apply that inverse to our measurement Y to directly solve for the state X. The interpretation is that the frequentist viewpoint treats estimation primarily as an inversion problem.

For a scenario with repeated, additive scalar measurements, we write

$$Y_n = X + e_n, \qquad n = 1, \ldots, N.$$

In words, the n-th measurement Y_n is equal to the true state X plus the n-th measurement error e_n, for indices n from 1 up to N. In this basic frequentist setup, our estimator is the sample mean: The interpretation is that repeated independent measurements can be averaged to reduce noise.

$$\hat{Y} = \frac{1}{N} \sum_{n=1}^{N} Y_n.$$

Expressed verbally, the estimate $\hat{Y}$ is one over the total number of samples N, multiplied by the sum from $n = 1$ to N of the individual measurements Y_n.

The purpose of this approach is to recover the state by direct inversion or averaging when the forward map and noise model are simple enough.

7.3. The Bayesian Approach and Likelihood Modeling

We shift from a direct inversion to a probabilistic framework incorporating prior knowledge and observation noise models; we do this because complex inverse problems are often ill-posed or non-invertible, and representing our knowledge as probability distributions allows for a mathematically rigorous quantification of uncertainty.

When G is not easily invertible, we transition to the "Bayesian approach," which the notes break down into two distinct steps.

Step "1) model for X yields prob dist P_X." In words, our mathematical understanding or assumptions about the underlying state X provide us with a prior probability distribution, denoted mathematically as P_X. The interpretation is that prior information is encoded probabilistically before any new data are incorporated.

Step "2) dist of measurement error." We must quantify the probability of observing a specific measurement given a known state. This is governed by the conditional probability equation:

$$P(Y \in dy \mid X = x) = e^{-\ell(x,y)}\, dy.$$

Reading this equation out loud: the probability that the measurement Y falls within an infinitesimal volume dy, given that the true state X is exactly x, is equal to the exponential of negative $\ell(x, y)$,

integrated over dy. The term $\ell(x, y)$ in the exponent is explicitly labeled in the notes as the "log likelihood."

The purpose of this likelihood model is to encode mathematically how observations are generated from states in the presence of measurement uncertainty.

7.4. The Posterior Distribution

We combine our initial prior model with the new measurement data using Bayes' theorem; we do this for the purpose of updating our mathematical belief about the true state of the system, culminating in the posterior distribution which serves as the complete solution to the Bayesian inverse problem.

By combining the prior and the likelihood, the notes state this "yields posterior." The fundamental equation for the posterior measure is

$$P(X \in dx \mid Y = y) = \frac{e^{-\ell(x,y)} P_X(dx)}{\int e^{-\ell(x,y)} P_X(dx)}.$$

To speak this equation: the posterior probability that the state X lies in the infinitesimal region dx, given that the measurement Y is exactly y, is a fraction. The numerator is the exponential of the negative log-likelihood $\ell(x, y)$ multiplied by the prior probability measure $P_X(dx)$. The notes annotate this numerator component involving the likelihood as the "Measurement density." The denominator is the integral over the entire state space of the same expression, $e^{-\ell(x,y)} P_X(dx)$. The notes contain a small annotation next to the denominator stating it acts to "norm to 1," ensuring the resulting posterior is a valid probability measure that integrates to one. The interpretation is that Bayes' rule reweights the prior by how compatible each state is with the observation and then renormalizes the result.

The purpose of the posterior is to give the fully updated probabilistic description of the unknown state after the data have been taken into account.

7.5. Graphical Intuition of the Bayesian Update

We visually represent the multiplication of probabilities; we do this to build a concrete geometric understanding of how a sharp, possibly multi-modal prior belief is modulated and shifted by the broader distribution of incoming measurement data.

At the bottom of the page, I have sketched three graphs to illustrate this multiplicative update.

The top-left graph is labeled "prior." The horizontal axis represents the state space, and the vertical axis represents probability density. The sketch shows two distinct, sharp peaks, indicating a strong prior belief that the state exists in one of two specific, narrow configurations.

There is a plus sign leading to the top-right graph, labeled "measurement density." This graph displays a single, smooth, broad curve—similar to a wide Gaussian distribution—representing the information provided by the noisy observation Y.

The bottom graph illustrates the resulting posterior distribution, which is the product of the prior and the measurement density, subsequently normalized. The sketch demonstrates that the posterior retains the sharp, multi-modal characteristics of the prior peaks, but their relative heights and exact shapes are now modulated and scaled by the broad envelope of the measurement density curve.

The purpose of these sketches is to show visually how prior information and data combine to produce the posterior distribution.

8. Continuous-Time Filtering and the Kallianpur–Striebel Formula

Let us continue our lecture series. We now turn to the fifth page of the Lecture 14 notes, where the continuous-time filtering problem is formulated and the Kallianpur–Striebel formula is introduced.

8.1. The Signal and Measurement Equations

We begin by establishing the governing stochastic differential equations for the system. The first equation defines the hidden signal, marked (S) in the notes:

$$dX_t = B(X_t)\,dt + C(X_t)\,dW_t.$$

Expressed in words, the differential of the state vector X_t is equal to the drift vector field $B(X_t)$ multiplied by dt, plus the diffusion matrix $C(X_t)$ integrated against the Wiener process dW_t. The state X_t evolves in $\mathbb{R}^d$. As noted in the margin, one assumes global Lipschitz conditions on the coefficients to guarantee existence and uniqueness of the solution. The interpretation is that X_t is the hidden dynamical system we want to track, with deterministic drift and random forcing.

The second equation governs the observation, labeled (O):

$$dY_t = H(X_t)\,dt + R\,dV_t, \qquad Y_0 = 0.$$

This says that the differential of the observation vector Y_t is equal to the observation function $H(X_t)$ multiplied by dt, plus the noise coefficient matrix R integrated against a separate Wiener process dV_t. The observation process takes values in $\mathbb{R}^p$. The interpretation is that the data consist of a noisy, lower-dimensional view of the hidden signal rather than a direct observation of the full state.

The notes stress two structural features of the problem. First, the true state X is not observed directly, so its initial condition is not available to us. Second, the observation dimension satisfies $p \ll d$, which reflects the fact that the data are much lower-dimensional than the hidden state.

Finally, we formalize the history of the observations by

$$\mathcal{Y}_{0:t} = \sigma(Y_s : s \le t).$$

In words, $\mathcal{Y}_{0:t}$ is the sigma-algebra generated by the observation process up to time t. It represents the information available up to time t. The purpose of this notation is to specify exactly what data the filter may use. The interpretation is that every estimator at time t must be measurable with respect to $\mathcal{Y}_{0:t}$, since only these observations are actually known.

8.2. Formulation of the Filtering Problem

With the system defined, we state the filtering problem:

$$\pi_t(A) = \mathbb{P}(X_t \in A \mid \mathcal{Y}_{0:t}), \qquad A \in \mathcal{B}(\mathbb{R}^d).$$

In words, $\pi_t(A)$ is the conditional probability that the hidden state X_t lies in the set A, given the observation history up to time t. The purpose of this definition is to formulate the optimal estimate of the hidden signal using only the data that have actually been observed. The interpretation is that π_t is the full probabilistic description of what we know about the hidden state after seeing the data up to time t.

8.3. The Kallianpur–Striebel Formula

To solve this problem, we introduce the central theorem of this section: the Kallianpur–Striebel formula. We let $(\bar{X}_t)$ be a copy of (X_t) such that $(\bar{X}_t)$ is independent of (X_t, V_t). In words, we construct an independent copy of the signal process and use it as an auxiliary variable in the formula. The purpose of this construction is to decouple the signal from the observations and rewrite the conditional law in a tractable form.

The theorem gives

$$\pi_t(A) := \mathbb{E}\big(\mathbf{1}_A(X_t) \mid \mathcal{Y}_{0:t}\big) = \frac{\mathbb{E}_{\bar{X}}\big(\mathbf{1}_A(\bar{X}_t) Z_t(\bar{X}, Y)\big)}{\mathbb{E}_{\bar{X}}\big(Z_t(\bar{X}, Y)\big)}.$$

This says that the conditional probability $\pi_t(A)$ is expressed as a ratio of two expectations taken only with respect to the independent copy $\bar{X}$, while the observed path Y is kept fixed. The purpose of the formula is to convert a conditional expectation into a normalized weighted expectation that is more accessible analytically and numerically. The interpretation is that paths of the independent copy $\bar{X}$ are reweighted by how well they explain the observed data, and the normalization turns these weights into a probability law.

The density process Z_t is defined by

$$Z_t(\bar{X}, Y) = \exp\left(\int_0^t \big\langle R^{-1} H(\bar{X}_s), dY_s \big\rangle - \frac{1}{2}\int_0^t \big\langle R^{-1} H(\bar{X}_s), H(\bar{X}_s) \big\rangle \, ds\right).$$

In words, $Z_t(\bar{X}, Y)$ is the exponential of a stochastic integral against the observation process together with a compensating quadratic term. The purpose of this density is to encode the likelihood weight of the observed data under the path $\bar{X}$. The interpretation is that $Z_t(\bar{X}, Y)$ scores how compatible a candidate signal path is with the observed trajectory Y.

8.4. Setting Up the Proof

At the bottom of the page, the proof begins by fixing a realization of the signal path $X(\omega)$. Once the signal path is frozen, the observation equation becomes a deterministic signal plus additive Gaussian noise. This is the point at which one can derive the conditional distribution of the observations given the state and apply change-of-measure arguments such as Girsanov's theorem. The interpretation is that one first simplifies the problem by treating the hidden path as known, and only afterward averages over all possible hidden paths.

9. Likelihood Densities, Measure Change, and the Zakai Equation

Let us resume our lecture. We now turn to the sixth page of the notes, which concludes the derivation of the continuous-time filtering problem and connects the probabilistic formulation to stochastic partial differential equations.

9.1. The Likelihood Density and the Radon–Nikodym Derivative

We continue the proof by computing the conditional law of the observations when the signal path is held fixed. In the notes, X is marked as "a realization," meaning that for this step we regard the hidden path as given. The relevant object is the Radon–Nikodym derivative of the conditional

observation measure with respect to the pure noise measure. The formula is written in the notes as

$$\frac{dP_{Y|X}}{dP_V} = \exp\left(\int_0^t \left\langle R^{-1}H(\bar{X}_s), dV_s\right\rangle - \frac{1}{2}\int_0^t \left\langle R^{-1}H(\bar{X}_s), H(\bar{X}_s)\right\rangle ds\right).$$

In words, the Radon–Nikodym derivative of the conditional law of the observations with respect to the Wiener measure P_V is given by an exponential weight involving a stochastic integral and its compensating quadratic term. The purpose of this formula is to quantify how much more likely the observed path is under a given hidden trajectory than under pure noise alone. The interpretation is that this exponential weight plays the role of a likelihood in path space.

The notes also stress an important distinction from the finite-dimensional Bayesian formula. One may write heuristically $e^{-\ell(x,y)}\,dy$, but here the correct object is not a Lebesgue density. It is a Radon–Nikodym density with respect to the Wiener measure P_V. The purpose of this distinction is to make clear that in path space the natural reference measure is a Gaussian process measure, not Lebesgue measure. The interpretation is that the likelihood must be defined relative to a probabilistic reference law for trajectories.

9.2. Independence and Measure Transformation

To justify the separation of state and observation noise, the notes invoke Girsanov's theorem. Under the corresponding reference measure, the observation drift is removed and the observation process behaves like pure noise, independent of the hidden signal. This yields the product structure

$$P_{(X,V)} = P_X \otimes P_V.$$

Reading this equation aloud: the joint probability law of the state X and the noise V is the tensor product of the signal law P_X and the noise law P_V. The purpose of this factorization is to justify why, in the Kallianpur–Striebel formula, one can integrate only over the independent copy of the state while keeping the observed path fixed. The interpretation is that the change of measure turns the coupled filtering problem into one involving independent signal and noise variables.

The notes also remark that the full proof should be checked in the script, since the complete measure-theoretic argument is longer than what fits into the handwritten summary.

9.3. The Unnormalized Measure and the SPDE Connection

The lower half of the page asks: what is the relevance to SPDEs? The bridge is the unnormalized conditional measure, denoted $\tilde{\eta}_t$. For a test function f, it is defined by

$$\tilde{\eta}_t(f) = \int f\, d\tilde{\eta}_t = \mathbb{E}_{\bar{X}}\big[f(\bar{X}_t)Z_t(\bar{X}, Y)\big].$$

To express this verbally, the unnormalized measure $\tilde{\eta}_t$ applied to a test function f is the expectation, taken only over the independent path $\bar{X}$, of the product of $f(\bar{X}_t)$ and the likelihood weight $Z_t(\bar{X}, Y)$. This is exactly the numerator of the Kallianpur–Striebel formula. The purpose of introducing $\tilde{\eta}_t$ is to remove the nonlinear normalization in Bayes' rule and thereby obtain a simpler linear evolution equation. The interpretation is that $\tilde{\eta}_t$ carries the same filtering information as π_t, but in a form better suited to stochastic calculus.

9.4. The Zakai Equation and Stochastic Calculus

To derive the dynamics of $\tilde{\eta}_t(f)$, one differentiates the product $f(\bar{X}_t)Z_t(\bar{X}, Y)$. The notes emphasize that this requires semimartingale theory, since both the state process and the likelihood process are continuous semimartingales. Ordinary calculus does not apply, so one must use the Itô product rule.

Applying the Itô product rule to $df(\bar{X}_t)$ and dZ_t, and then taking the expectation $\mathbb{E}_{\bar{X}}$, leads to a linear stochastic partial differential equation for the unnormalized filter. As indicated in the notes, this SPDE is the Zakai equation. The interpretation is that the filtering problem is converted into the evolution of an unnormalized conditional density governed by a linear SPDE.

Once $\tilde{\eta}_t$ is known, the normalized filter is recovered by dividing by the total mass:

$$\pi_t(f) = \frac{\tilde{\eta}_t(f)}{\tilde{\eta}_t(1)}.$$

In words, the true conditional estimate is obtained by normalizing the unnormalized measure at time t. The purpose of this final step is to pass back from the linear Zakai equation to the actual conditional law of the hidden state. The interpretation is that normalization restores the probabilistic meaning after the linear SPDE has been solved.

Appendix A

A.1. Difference Between P_{t-s} and $P_t(x-y)$

The notation P_{t-s} refers to the heat semigroup as an operator acting on a function, while $P_t(x-y)$ refers to the corresponding heat kernel evaluated at the spatial points x and y. These are two ways of describing the same object. More precisely, if ϕ is a test function, then

$$P_{t-s}\phi(x) = \int P_{t-s}(x-y)\phi(y)\,dy.$$

Thus, P_{t-s} emphasizes the operator that evolves data from time s to time t, whereas $P_{t-s}(x-y)$ emphasizes the kernel representation of that operator in space. The purpose of this distinction is to clarify when we are viewing the heat semigroup as an operator and when we are viewing it through its kernel.

A.2. Breakdown of $M^{(1)}_{T,X}v(t,x)$

Recall that

$$M^{(1)}_{T,X}v(t,x) = \int_0^t e^{-(t-s)}P_{t-s}\big(g(u(s,\cdot))\partial_x v(s,\cdot) + f(u(s,\cdot))\big)(x)\,ds.$$

Its pieces can be displayed as

$s \in [0,t]$	past time variable,
$e^{-(t-s)}P_{t-s}$	damped heat semigroup from time s to time t,
$g(u(s,\cdot))\partial_x v(s,\cdot)$	transport term,
$f(u(s,\cdot))$	drift term,
$(\cdot)(x)$	evaluation at the spatial point x.

The mild form comes from Duhamel's principle:

solution at time t = linear evolution of initial data+$\int_0^t$ semigroup from s to $t\times$forcing at time $s\,ds$.

In the present setting, the forcing is

$$g(u(s,\cdot))\partial_x v(s,\cdot) + f(u(s,\cdot)),$$

so the deterministic contribution is exactly

$$M^{(1)}_{T,X} v(t,x) = \int_0^t e^{-(t-s)} P_{t-s}\big(g(u(s,\cdot))\partial_x v(s,\cdot) + f(u(s,\cdot))\big)(x)\,ds.$$

The purpose of this appendix note is to make clear which deterministic terms are gathered into $M^{(1)}_{T,X}$.

A.3. Breakdown of $M^{(2)}_{T,X} v(t,x)$

Recall that

$$M^{(2)}_{T,X} v(t,x) = \int_0^t \int_0^{2\pi} e^{-(t-s)} P_{t-s}(x-y) g(u(s,y))\,dX_s(y)\,ds.$$

Its pieces can be displayed as

$s \in [0,t]$	past time variable,
$y \in [0,2\pi]$	spatial integration variable,
$e^{-(t-s)} P_{t-s}(x-y)$	damped heat semigroup kernel from (s,y) to (t,x),
$g(u(s,y))$	coefficient multiplying the rough forcing,
$dX_s(y)$	integration against the spatial rough signal at time s,
$\int_0^{2\pi} \cdots dX_s(y)$	rough spatial integral at fixed time s,
$\int_0^t \cdots ds$	accumulation over all past times s.

The mild form again comes from Duhamel's principle, but now the forcing term is rough in space. At each fixed time s, one first forms the rough spatial integral

$$\int_0^{2\pi} P_{t-s}(x-y) g(u(s,y))\,dX_s(y),$$

and then propagates that contribution to time t with the damping factor $e^{-(t-s)}$. Hence the rough contribution is

$$M^{(2)}_{T,X} v(t,x) = \int_0^t \int_0^{2\pi} e^{-(t-s)} P_{t-s}(x-y) g(u(s,y))\,dX_s(y)\,ds.$$

The purpose of this appendix note is to make clear how the rough forcing enters into $M^{(2)}_{T,X}$.

A.4. Meaning of $|||Y|||_{\mathbb{X},\alpha}$

The quantity $|||Y|||_{\mathbb{X},\alpha}$ denotes the controlled rough path norm of Y relative to the reference rough path $\mathbb{X}$ with Hölder exponent α. In rough path theory, this means that Y is not viewed in isolation: it is viewed together with a description of how its increments are controlled by the increments of the rough path $\mathbb{X}$. Thus, $|||Y|||_{\mathbb{X},\alpha}$ measures both the Hölder size of Y and its compatibility with the rough structure generated by $\mathbb{X}$. The purpose of this norm is to measure both the size of Y and its compatibility with the reference rough path.

A.4.1. Controlled Rough Path Norm

More concretely, if Y is controlled by X with Gubinelli derivative Y', so that the remainder

$$R^Y(x,y) := Y(y) - Y(x) - Y'(x)\big(X(y) - X(x)\big)$$

has better regularity, then a typical controlled rough path norm is

$$|||Y|||_{\mathbb{X},\alpha} := \|Y\|_{C^\alpha} + \|Y'\|_{C^\alpha} + \|R^Y\|_{C^{2\alpha}}.$$

Here,

$$\begin{aligned}
\|Y\|_{C^\alpha} &:= \sup_{x\neq y} \frac{|Y(y) - Y(x)|}{|y-x|^\alpha}, \\
\|Y'\|_{C^\alpha} &:= \sup_{x\neq y} \frac{|Y'(y) - Y'(x)|}{|y-x|^\alpha}, \\
\|R^Y\|_{C^{2\alpha}} &:= \sup_{x\neq y} \frac{|R^Y(x,y)|}{|y-x|^{2\alpha}}.
\end{aligned}$$

This formula records precisely that the increment of Y is approximated to first order by $Y'(x)(X(y)-X(x))$, with a remainder that is of higher order. The purpose of this definition is to quantify the controlled expansion of Y relative to X.

A.4.2. Rough Path Norm of $\mathbb{X}$

The rough path norm of $\mathbb{X}$ itself is typically written as

$$|||\mathbb{X}|||_\alpha := \|X\|_{C^\alpha} + \|\mathbb{X}\|_{C^{2\alpha}},$$

where

$$\|\mathbb{X}\|_{C^{2\alpha}} := \sup_{x\neq y} \frac{|\mathbb{X}(x,y)|}{|y-x|^{2\alpha}}.$$

Thus, $|||\mathbb{X}|||_\alpha$ measures both the α-Hölder size of the first-level path X and the 2α-Hölder size of its second-order component $\mathbb{X}$. The purpose of this norm is to record both the first-level rough path and the second-order information needed for rough integration.

A.5. Hölder Seminorm and Norm

Strictly speaking, the quantity

$$[Y]_{C^\alpha} := \sup_{x\neq y} \frac{|Y(y) - Y(x)|}{|y-x|^\alpha}$$

is a Hölder seminorm rather than a true norm, because it vanishes on every constant function. A genuine Hölder norm is obtained by adding the uniform size:

$$\|Y\|_{C^\alpha} := \|Y\|_{L^\infty} + [Y]_{C^\alpha}.$$

Here,

$$\|Y\|_{L^\infty} := \sup_x |Y(x)|$$

is the supremum norm, which measures the maximum absolute size of the function Y over its domain. The purpose of this distinction is to separate pure regularity control from full size control.

A.6. Spatial Variable Instead of Time

In classical stochastic integration, the irregular driving signal is usually indexed by time, for example in an integral of the form

$$\int_0^t H_s \, dW_s.$$

Here, the integration variable is the time variable s. In the present rough path setting, however, the irregular driving signal is indexed by space, as in

$$\int_0^{2\pi} Z_y \, dX(y).$$

Here, the integration variable is the spatial variable y. Thus, when we say that we are replacing the temporal variable by a spatial variable, we mean that the rough path integration is performed along space rather than along time. The purpose of this note is to emphasize that the rough integration parameter here is spatial rather than temporal.

A.7. Derivation of the Bound in 4.3

Let

$$I_t := \int_0^t (Z_y - Z_0) \, dX_y.$$

To estimate the α-Hölder norm of I, we look at an increment over an interval $[s, t]$:

$$I_t - I_s = \int_s^t (Z_y - Z_s) \, dX_y + (Z_s - Z_0)(X_t - X_s).$$

Now assume that Z is controlled by X, so that

$$Z_y - Z_x = Z'_x(X_y - X_x) + R_{x,y}.$$

Applying the rough integral expansion on $[s, t]$ gives

$$\int_s^t (Z_y - Z_s) \, dX_y = Z'_s \, \mathbb{X}(s, t) + \mathrm{Error}_{s,t},$$

where the error is controlled by the remainder term and the Hölder regularity of Z'. More precisely,

$$|\mathrm{Error}_{s,t}| \le C\Big(\|X\|_\alpha \|R\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Z'\|_\alpha\Big) |t - s|^{3\alpha}.$$

Since

$$|(Z_s - Z_0)(X_t - X_s)| \le \|Z\|_{C^\alpha}\|X\|_{C^\alpha}|t-s|^\alpha,$$

the full increment of I is therefore controlled by the same rough path data. Dividing by $|t-s|^\alpha$ and taking the supremum over $s \neq t$ yields

$$\left\| \int_0^{\cdot} (Z_y - Z_0)\, dX_y \right\|_\alpha \le C\Big(\|X\|_\alpha \|R\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha}\|Z'\|_\alpha \Big).$$

Thus, the first term on the right-hand side comes from the remainder $R_{x,y}$, while the second term comes from the second-order component $\mathbb{X}$ together with the Gubinelli derivative Z'. The purpose of this estimate is to separate the contribution of the remainder from the contribution of the second-order rough term.

A.8. Why the Expansion on $[s,t]$ Has the Form $Z'_s\mathbb{X}(s,t) + \mathrm{Error}_{s,t}$

If Z is controlled by X, then for $y \in [s,t]$ one has

$$Z_y - Z_s = Z'_s(X_y - X_s) + R_{s,y}.$$

Integrating this identity against dX_y over the interval $[s,t]$ gives

$$\int_s^t (Z_y - Z_s)\, dX_y = \int_s^t Z'_s(X_y - X_s)\, dX_y + \int_s^t R_{s,y}\, dX_y.$$

Since Z'_s is frozen at the left endpoint s, it can be pulled out of the first integral, and the remaining term is exactly the second-order rough path increment:

$$\int_s^t Z'_s(X_y - X_s)\, dX_y = Z'_s\, \mathbb{X}(s,t).$$

The second integral,

$$\int_s^t R_{s,y}\, dX_y,$$

is of higher order and is therefore absorbed into the error term $\mathrm{Error}_{s,t}$. This is why the rough integral expansion takes the form

$$\int_s^t (Z_y - Z_s)\, dX_y = Z'_s\, \mathbb{X}(s,t) + \mathrm{Error}_{s,t}.$$

The purpose of this expansion is to identify the leading rough term before estimating the higher-order error.

A.9. Derivation of the Error Bound

Starting again from the controlled expansion

$$Z_y - Z_s = Z'_s(X_y - X_s) + R_{s,y},$$

we integrate against dX_y on the interval $[s,t]$:

$$\int_s^t (Z_y - Z_s)\, dX_y = \int_s^t Z'_s(X_y - X_s)\, dX_y + \int_s^t R_{s,y}\, dX_y.$$

The first term is the leading rough path contribution,

$$\int_s^t Z'_s(X_y - X_s)\, dX_y = Z'_s\, \mathbb{X}(s,t).$$

Hence the error term is what remains after subtracting this leading part. The rough sewing estimate shows that

$$\left| \int_s^t (Z_y - Z_s)\, dX_y - Z'_s \mathbb{X}(s,t) \right| \le C\Big(\sup_{y\in[s,t]} |R_{s,y}|\, |X_{s,t}| + \sup_{y\in[s,t]} |Z'_y - Z'_s|\, |\mathbb{X}(s,t)| \Big).$$

Now use the Hölder bounds

$$|R_{s,y}| \le \|R\|_{2\alpha} |y-s|^{2\alpha}, \qquad |X_{s,t}| \le \|X\|_\alpha |t-s|^\alpha,$$

and

$$|Z'_y - Z'_s| \le \|Z'\|_\alpha |y-s|^\alpha, \qquad |\mathbb{X}(s,t)| \le \|\mathbb{X}\|_{2\alpha} |t-s|^{2\alpha}.$$

Since $|y-s| \le |t-s|$ for $y \in [s,t]$, it follows that

$$\sup_{y\in[s,t]} |R_{s,y}|\, |X_{s,t}| \le \|X\|_\alpha \|R\|_{2\alpha} |t-s|^{3\alpha},$$

and

$$\sup_{y\in[s,t]} |Z'_y - Z'_s|\, |\mathbb{X}(s,t)| \le \|\mathbb{X}\|_{2\alpha} \|Z'\|_\alpha |t-s|^{3\alpha}.$$

Combining these two bounds yields

$$|\mathrm{Error}_{s,t}| \le C\Big(\|X\|_\alpha \|R\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Z'\|_\alpha \Big) |t-s|^{3\alpha}.$$

Thus, one part of the error comes from the remainder term $R_{s,y}$, while the other comes from the Hölder regularity of the Gubinelli derivative Z'.